

# Homework 1

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**Problem 1 part 1**

A matrix  $A$  is invertible if  $\det(A) \neq 0$ . To show that a monomial matrix is invertible, the same property must apply. To prove that a monomial matrix is invertible, we will take the determinant of the base case, a  $2 \times 2$  matrix, and the inductive case, a  $k \times k$  matrix with  $k > 2$ , are both non zero.

Say  $A \in \mathbb{R}^{2 \times 2}$  is a monomial matrix. The determinant of  $A$  is  $ad - cb$ , where the variables come from  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . Since the matrix is monomial, either  $ac$  or  $-bd$  evaluates zero, as there can only be one non-zero value per column and row in the matrix. Therefore,  $\det(A) \neq 0$ .

For the inductive step, where  $A \in \mathbb{R}^{k \times k}$ , the determinant can be calculated in the same way as a diagonal matrix, the product of all non-zero entries of the matrix. A monomial matrix requires that  $A$  only has one non-zero entry per row and column, the same as a diagonal matrix. To convert a monomial matrix to a diagonal matrix, the rows of the matrix need to be rearranged where  $A_{ii} \neq 0$ . In a  $2 \times 2$  monomial matrix, rearranging the rows affects the determinant as follows,  $bd - ac$ . The transformation is just multiplying the determinant by  $-1$ . If we apply another rearranging of the rows, the determinant goes back to its original formulation, essentially multiplying the determinant by  $-1^2$ . Applying this to a general case, if rearranging  $l$  rows of a matrix results in the determinant being multiplied by  $(-1)^l$ , then the determinant of a monomial matrix is, after rearranging the rows to make it diagonal,  $(-1)^l \prod_{i=1}^n (A_{ii})$ , where  $l$  is the number of row swaps being made. Therefore, for a  $k \times k$  monomial matrix, the determinant is non-zero. By proof of induction, a monomial matrix is invertible.

**Problem 1 part 2**

Given  $A$  is skew symmetric,  $A^T = -A$ . In quadratic form,  $x^T A x = 0$ . By transposing the entire expression,  $(x^T A x = 0)^T = x A^T x^T = x(-A)x^T$ . The resulting expression is  $x^T A x = -x A x^T$ , rewriting it as  $2x^T A x = 0$  and  $x^T A x = 0$ . Therefore  $x^T A x = 0$ .

**Problem 2 part 1**

$P_2 E_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ l_{41} & 0 & 0 & 1 \\ l_{21} & 1 & 0 & 0 \\ l_{31} & 0 & 1 & 0 \end{pmatrix}$ .  $P_2^{-1} = P_2$  since the gaussian elimination only requires the swapping of rows 4, 2 and 3 on the identity matrix.  $P_2 E_1 P_2^{-1} = P_2 E_1 P_2 = \tilde{E}_1$ .

$$\tilde{E}_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ l_{41} & 0 & 1 & 0 \\ l_{21} & 0 & 0 & 1 \\ l_{31} & 1 & 0 & 0 \end{pmatrix}$$

**Problem 2 part 2**

$\tilde{E}_k$  is different from  $E_k$  by the fact that  $\tilde{E}_k$  has both its rows  $i, j$  and columns  $i, j$  permuted. As seen by the first part of this problem, when  $E_k$  is multiplied by the permutation matrix from the left, only the rows of the  $E_k$  are permuted. If it is multiplied from its right, its columns are permuted. Since  $\tilde{E}_k$  needs to equal  $E_k$  with its rows permuted,  $\tilde{E}_k$  needs to have both its rows and columns permuted to satisfy the equation.

**Problem 3 part 1**

In order to get  $A = LU$ , the following row operations need to be executed.

1.  $R_2 = R_2 - (2R_1)$
2.  $R_3 = R_3 - (\frac{3}{2}R_2)$

This results in  $L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & \frac{3}{2} & 1 \end{pmatrix}$  and  $U = \begin{pmatrix} 6 & 10 & 0 \\ 0 & 6 & 4 \\ 0 & 0 & 6 \end{pmatrix}$ .

To solve for  $Ly = b$ , we use forward substitution, where  $y_1 = 4$ ,  $y_2 = 2$ ,  $y_3 = -6$ . Then, to solve  $Ux = y$ , we use backwards substitution, where  $x_3 = -1$ ,  $x_2 = 1$ ,  $x_1 = -1$ .

**Problem 3 part 2**

$$A = \begin{pmatrix} 1 & -5 & 1 \\ 10 & 0 & 20 \\ 5 & 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 7 \\ 6 \\ 4 \end{pmatrix}$$

In order to solve this, we need to create a permutation matrix  $P$  by swapping the first row with the second row of  $A$ .

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \tilde{A} = \begin{pmatrix} 10 & 0 & 20 \\ 1 & -5 & 1 \\ 5 & 0 & -1 \end{pmatrix}, \quad \tilde{b} = \begin{pmatrix} 6 \\ 7 \\ 4 \end{pmatrix}$$

Now we can use row operations to obtain  $PA = LU$ . The row operations are as follows.

$$1. \ R_2 = R_2 - (\tfrac{1}{10}R_1)$$

$$2. \ R_3 = R_3 - (\tfrac{1}{2}R_1)$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{10} & 1 & 0 \\ \frac{1}{2} & 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 10 & 0 & 20 \\ 0 & -5 & -1 \\ 0 & 0 & -11 \end{pmatrix}.$$

With both triangular matrices, we can solve  $Ly = b$ , where  $y_1 = 6$ ,  $y_2 = \frac{32}{5}$ ,  $y_3 = 1$ . Solving  $Ux = y$  gives  $x_3 = \frac{-1}{11}$ ,  $x_2 = \frac{-347}{275}$ ,  $x_1 = \frac{86}{110}$ .

$$x = \begin{pmatrix} \frac{86}{110} \\ \frac{-347}{275} \\ \frac{-1}{11} \end{pmatrix}$$

**Problem 4**